Autoresearch findings

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Abstract

This report collects the findings: An unexploitable modal agent that defects against itself; When graded PrudentBots cooperate with graded FairBots; Families of unexploitable agents that defect against themselves; A certified map of variants of Gödel's ontological argument; A census of modal agents in the prisoner's dilemma; Settling informal claims about modal agents in Chicken; Three-player modal agents in a public-goods game.

1 An unexploitable modal agent that defects against itself

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent in Peano arithmetic, written with the provability box $\Box$ of the logic GL. FairBot, $\Box p$, cooperates exactly when it can prove that the opponent cooperates with it (p); CooperateBot always cooperates; DefectBot always defects. An agent is *unexploitable* if no modal opponent ever gets it to cooperate while the opponent defects. We consider agents built from p, from q ("the opponent cooperates with DefectBot"), $\bot$, $\neg$, $\wedge$, $\vee$ and $\Box$, with every atom under a box, and measure size as the number of nodes in the formula.

Result

The agent $W = \Box(\Box\Box\bot \vee (p \wedge (q \vee \neg\Box q)))$ is unexploitable, cooperates with FairBot and with CooperateBot, yet defects against an exact copy of itself. An exhaustive search shows that no agent of size 11 or less has this combination of properties, and W (size 12) is one of the three smallest. It is the first member of an infinite family $W_n = \Box(\Box^n\bot \vee (p \wedge (q \vee \neg\Box q)))$, $n \geq 2$, all with the same four properties, with exact laws for how W_n plays against graded FairBots $\Box^k p$, graded PrudentBots and other members of the family.

Why it matters

Every unexploitable agent in the literature we checked (FairBot, PrudentBot, CliqueBot) cooperates with its own copy. W_n shows that unexploitability, cooperation with FairBot and generosity towards CooperateBot do not force self-cooperation: the delayed "gate" $\Box^n \bot$ lets the agent tolerate FairBot's failure to prove anything about DefectBot, while two copies of W_n never reach mutual cooperation.

Proved vs checked

The properties of W_n and its laws against graded FairBots, graded PrudentBots and other W_m are proved for all indices in Lean 4 (height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [17], cited, not formalised; see also [8]). The minimality of size 12 is an exhaustive computation over all agents up to size 12 in the stated language.

We believe this result is new. Our prior-art search (2026-10-17) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" (Alignment Forum).
- Mennen, "Modal Combat for games other than the prisoner's dilemma" (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the proof and check files are included so anyone can re-run them.

Claim K6

In L (atoms p,q boxed; F,~,&,|,[]; size=nodes), the smallest agent that cooperates with FairBot and CooperateBot, defects against itself and is unexploitable has size 12; W=[]([][]F|(p&(q|~[]q))) is one, unexploitable by every opponent; none of size <=11.

Status: derived.

Verified.

- `checks/K6-c046_combo12.py`: census counts to size 12; S=0,FBc=1,CBc=1 counts per size 4..12 = 2,2,4,8,19,46,120,357,900; none <=11 and exactly 3 of size 12 unexploited by all 2815 size<=9 classes; W's self/FB/CB/DB outcomes and zero exploiters by an independent height-table solver; GLS certificates for W vs W, FB, CB with flipped-candidate controls — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K9

For every n>=2, the agent W_n=[]([]^n F|(p&(q|~[]q))) of size n+10 is unexploitable by every modal opponent, mutually cooperates with FairBot and CooperateBot, and defects against itself (both copies defect above height n+1); W_0, W_1 fail with FairBot. Lean-proved.

Status: derived.

Verified.

- `checks/K9-c046_wn_lean.py`: embedded Lean 4 theorems w_gate, w_unexploitable, w_db, w_cb, w_self, w_self_db, w_fb compile with no sorryAx; three false variants rejected; height tables n<=12 and 3000 random opponent traces agree; size(W_n)=n+10 — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K11

For all n>=0, k>=1, j,m>=0 (height semantics of closed GL fixed points): W_n and F_k cooperate at every height iff k<n; W_n and P_j iff 2<=n<j; W_n and W_m both defect from height n+m+3. Hence all W_n (n>=2) are distinct. Lean-proved.

Status: derived.

Verified.

- `checks/K11-c046_wpairs_lean.py`: embedded Lean 4 theorems w_fk, pj_db, w_pj, w_wm compile with no sorryAx; four false variants rejected with proof errors; height tables n<=12, k,j,m<=14 (heights 0..79) agree with all theorems and K10's n=2..5 predicates — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261017-1126`, claims K6, K9, K11, run-folder commit `662166ecff0651902cc5fcb04bf747c62d795af2`, published 2026-10-18.

2 When graded PrudentBots cooperate with graded FairBots

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. A *modal agent* [1] decides by asking what Peano arithmetic (PA) can prove about the opponent, written with the provability box $\Box$ of the logic GL. Here p means "the opponent cooperates with me" and q means "the opponent cooperates with DefectBot", the agent that always defects. FairBot, $\Box p$, cooperates exactly when it can prove that the opponent cooperates with it. PrudentBot adds a safety test: it also requires a proof, in the stronger system PA+1, that the opponent would not cooperate with DefectBot, so it does not reward an opponent that cooperates with everyone.

We study two graded families. The graded FairBot $F_k = \Box^k p$ $(k \geq 1)$ asks for k nested layers of provability; F_1 is FairBot. The graded PrudentBot $P_j = \Box p \wedge \Box(\neg\Box^j \bot \to \neg q)$ $(j \geq 0)$ runs its safety test in PA+j, since $\neg\Box^j\bot$ expresses the consistency of PA+$(j-1)$; P_1 is PrudentBot.

Result

P_j and F_k cooperate with each other exactly when $k \leq j$; when $k > j$, both defect. In words: a PrudentBot that runs its safety test in a stronger system can afford to cooperate with FairBots that demand more nested proof, and the exchange rate is one for one. Every P_j still defects against CooperateBot, so the extra cooperation costs nothing in safety.

Why it matters

Barasz et al. remark that PrudentBot's safety test may use a stronger system than PA+1 and that this widens the set of agents it cooperates with, and online discussion of the paper looked at specific pairs of such variants. Here the relation

is pinned down exactly for the two graded families, for all indices: the boundary of cooperation is the single inequality $k \leq j$.

Proved vs checked

Proved for all $j \geq 0$ and $k \geq 1$ in Lean 4, in the height semantics of closed GL fixed points (on the chain of heights, both agents cooperate at every height when $k \leq j$; when $k > j$ both defect above height $j + k$). The link to provability in PA is Solovay's theorem [17] (see also [8]), cited, not formalised, and the reading in the standard model assumes PA is sound.

This result extends # Prior-art answer for R4 (K8: P_j / F_k threshold law)

Verdict: partial prior art.

- Barasz et al. 2014, remark after Theorem 3.2: PrudentBot's DefectBot check may use PA+n for larger n, which widens the set of agents it cooperates with without losing safety. That is the idea behind graded PrudentBots, stated informally.
- LessWrong comments on the Barasz et al. post (orthonormal, cousin_it) discuss indexed variants "Prudent(m,n)Bot" with specific pairs, e.g. Prudent(0,100)Bot cooperating with Prudent(0,1)Bot and Prudent(99,1000)Bot, and cousin_it noting that a PrudentBot using PA and PA+1 does not cooperate with one using PA+1 and PA+2.
- Not found: a general threshold law for P_j against graded FairBots F_k (cooperate iff k <= j, both defect above j+k), or a proof of it.
- Suggested wording: the graded family was discussed informally (Barasz et al. remark; LessWrong comments); the general law and its Lean proof were not found.

Sources searched (owner, web search, 2026-10-17)

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" (Alignment Forum).

- Mennen, "Modal Combat for games other than the prisoner's dilemma" (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

Not searched: Critch's parametric bounded Löb paper, theses, the contents of the Haskell agents folder, non-English sources. A "none" verdict means "not found in this search", nothing more. Our prior-art search (2026-10-17) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10 and the obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem (intelligence.org/files/ProgramEquilibrium.pdf).
- LessWrong post "Robust Cooperation in the Prisoner's Dilemma" and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" (Alignment Forum).
- Mennen, "Modal Combat for games other than the prisoner's dilemma" (agentfoundations.org item 1304).
- github.com/machine-intelligence/provability (Haskell solver and combat code; the agents folder contents were not visible).

If you know of related work we missed, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the proof and check files are included so anyone can re-run them.

Claim K8

For all j>=0, k>=1 (height-chain semantics of closed GL fixed points): P_j=[]p&[](~[]^jF->~q) and F_k=[]^k p cooperate at every height iff k<=j; if j<k both defect above height j+k. Lean-proved; P_j defects vs CB.

Status: derived.

Verified.

- `checks/K8-c046_pjfk_lean.py`: embedded Lean 4 proof of pj_fk and pj_cb compiles with no sorryAx; flipped-conclusion variant is rejected by Lean; explicit height table for j<=8,k<=8 matches (coop everywhere iff k<=j, else both defect above j+k with exact threshold) — pass (Lean-proved, no sorryAx)

Prior art. Partial prior art.

Provenance. Run `20261017-1126`, claims K8, run-folder commit `662166ecff0651902cc5fcb04bf747c62d79` published 2026-10-18.

3 Families of unexploitable agents that defect against themselves

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL. p means "the opponent cooperates with me" and q means "the opponent cooperates with DefectBot". FairBot is $\Box p$, graded FairBots are $F_k = \Box^k p$, graded PrudentBots are $P_j = \Box p \wedge \Box(\neg\Box^j\bot \to \neg q)$, and ConBot is $\neg\Box\bot$. An agent is *unexploitable* if no modal opponent ever gets it to cooperate while the opponent defects.

An earlier finding on this site described $W_n = \Box(\Box^n\bot \vee (p \wedge (q \vee \neg\Box q)))$: unexploitable, cooperative with FairBot and CooperateBot, yet defecting against its own copy. This finding describes four further families of unexploitable agents that defect against themselves, and how they relate.

Result

- **Gate families** $V_{n,r} = \Box(\Box^n\bot \vee (p \wedge (q \vee \neg\Box^r q)))$, which contain $W_n = V_{n,1}$. $V_{n,r}$ and F_k cooperate exactly when $k + r \le n$; $V_{n,r}$ and P_j exactly when $r < n < j$; any two members defect against each other. For $r < n$ every member is unexploitable, and these members are pairwise distinct.

- **Prudent families** $G_{m,j} = \Box(\Box^m p \wedge (q \vee \Box(\Box^j\bot \vee \neg q)))$ **and** $H_{m,j} = \Box^{m+1}(p \wedge (q \vee \Box(\Box^j\bot \vee \neg q)))$, $m \ge j \ge 1$. Both are unexploitable, cooperate with FairBot and CooperateBot, defect against themselves, and cooperate with F_k exactly when $k \le j$ and with P_i exactly when $i > m$. So $G_{m,j}$ and $H_{m,j}$ cannot be told apart by graded FairBots or PrudentBots; the agent $S_j = \Box^{j+1}(p \vee \neg\Box^{j+1} q)$ separates them ($H_{m,j}$ defects against it while it cooperates; against $G_{m,j}$ both defect). ConBot separates the two kinds of family: $G_{m,j}$ defects against it, every $V_{n,r}$ with $n \ge 1$ does not.

- **A common family** $J_{a,b,j} = \Box^{a+1}(\Box^b p \wedge (q \vee \Box(\Box^j\bot \vee \neg q)))$, which contains G ($a = 0$) and H ($b = 0$). Checked for $a, b \le 5$, $j \le 6$: $J_{a,b,j}$ cooperates with F_k exactly when $k \le j$ and with P_i exactly when $i > a+b$, and exact laws hold against a two-parameter family of separating probes; the 70 members with $j \le a + b \le 5$ are pairwise distinct. The size-13 member $J_{1,1,1}$ has the same outcomes as $G_{2,1}$ and $H_{2,1}$ against $F_1, \ldots, F_4$ and $P_0, \ldots, P_5$, yet two opponents of size at most 9 distinguish all three.

Why it matters

The unexploitable agents we found in the literature (FairBot, PrudentBot and its graded and probabilistic variants [1], [16]) all cooperate with their own copies. These families show that defecting against oneself is not a single odd example: it comes in several infinite families with different mechanisms (a delayed gate on the DefectBot test in V, a PrudentBot-style check in G, H and J). Their behaviour against graded FairBots and PrudentBots is described by simple thresholds, and these opponents alone do not identify an agent: distinct members can agree on all of them and need other probes to be told apart.

Proved vs checked

The laws for $V_{n,r}$, $G_{m,j}$ and $H_{m,j}$ (including the separations by ConBot and S_j) are proved for all indices in Lean 4, in the height semantics of closed GL fixed points; the link to provability in arithmetic is Solovay's theorem [17], cited, not formalised (see also [8]). The statements about $J_{a,b,j}$ and $J_{1,1,1}$ are checked, not proved: exact tables for the stated parameter ranges, with every member of size at most 15 independently certified by GL fixed-point solving. The unexploitability of $J_{a,b,j}$ is supported by a short argument (the agent implies $\Box^{a+b+1}p$) and randomised safety tests, not by a proof for all indices.

We believe this result is new. Our prior-art searches (2026-10-17 to 2026-10-18) covered:

- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: named agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10; obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem.
- LessWrong post on Barasz et al. and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling"; Garrabrant, "Modal SAT: Introduction" (2015).
- Mennen, "Modal Combat for games other than the prisoner's dilemma".
- Critch, Parametric Bounded Löb's Theorem and Robust Cooperation of Bounded Agents (arXiv 1602.04184), first sections.
- Searches for round-robin, tournament, replicator or ESS analyses of modal agents: only iterated-PD and FDT evolution papers (e.g. arXiv 2005.05154, no modal agents).

- github.com/machine-intelligence/provability (owner, 2026-10-18): its example agent file (target.ua) and the Haskell agent catalogue (Agents.hs) (CB, DB, FB, FB_1..5, TbF_1..3, FB_r, MB, WB, WB_2, aMB, sMB, IB, PB, PB_2, PB_3, NB, JB, TB, QB_1..3, modal UDT). Evaluated in height semantics (our reading of its [k] and boxk notation as PA+k provability): none of them defects against itself while cooperating with FairBot and CooperateBot; none has the form of V, G, H or J.
- Barasz et al. 2014 (arXiv 1401.5577), full text; LaVictoire et al. 2014 (Program Equilibrium via Löb's Theorem).
- LessWrong post on Barasz et al. and comments; Arbital "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" and "Modal SAT: Introduction"; Mennen, "Modal Combat for games other than the prisoner's dilemma".
- Critch, Parametric Bounded Löb (arXiv 1602.04184), first sections.
- Oesterheld 2022, A Note on the Compatibility of Different Robust Program Equilibria of the Prisoner's Dilemma (arXiv 2211.05057).

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was partly proved in Lean (claims K14, K18, K21) and partly checked by script (claims K20, K22, K23); the proof and check files are included so anyone can re-run them.

Claim K14

For all n>=0, r>=1, k>=1, j>=0 (height semantics of closed GL fixed points), V_{n,r}=[]([]^nF|(p&(q|~[]^r q))): CC with F_k iff k+r<=n; with P_j iff r<n<j; any two V's defect. For r<n each is universally unexploitable, all pairwise distinct. Lean.

Status: derived.

Verified.

- `checks/K14-c046_vfam_lean.py`: embedded Lean 4 theorems qclause_r, v_unexploitable, v_db, v_cb, v_fk, pj_db, v_pj, v_vm, v_params compile with no sorryAx; 5 false variants rejected; literal height tables n<=10, r<=5, k,j<=12 and V-vs-V n,m<=6, r,s<=3 agree; 35 r<n agents have distinct F/P outcome vectors; size(V_{n,r})=n+r+9 — pass (Lean-proved, no sorryAx)
- `checks/K14-c046_vfam_lean.py`: embedded Lean 4 theorems qclause_r, v_unexploitable, v_db, v_cb, v_fk, pj_db, v_pj, v_vm, v_params compile with no sorryAx; 5 false variants rejected; literal height tables agree;

35 r<n agents have distinct F/P outcome vectors — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K18

For all m>=j>=1 (height semantics of closed GL fixed points), G_{m,j}=[]([]^m p&(q|[]([]^jF|~q))), size m+j+10, is unexploitable, coops with FB and CB, defects vs itself; CC with F_k iff k<=j, P_i iff i>m; defects vs ~[]F unlike every V_{n,r} with n>=1. Lean.

Status: derived.

Verified.

- `checks/K18-c046_gfam_lean.py`: embedded Lean 4 theorems gq, g_unexploitable, g_db, g_cb, g_conbot, g_fk, g_pj, g_gg, v_conbot, g_params compile with no sorryAx; 4 false variants rejected; literal height tables m<=8, j<=9, k,i<=11 and G-vs-G m,m'<=5 agree; independent string solver: self/FB/CB/DB/ConBot for m,j<=3, W_2 coops with ConBot; size(G_{m,j})=m+j+10 — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K20

For all H_{m,j}=box^{m+1}(p&(q|box(box^j F|~q))), 1<=j<=m, size m+j+10<=15 (six agents), self-play is DD, FB/CB CC; CC with F_k iff k<=j (k<=4), P_i iff i>m (i<=5). GLS certifies 102 cells including separators from matching G and bounded V.

Status: derived.

Verified.

- `checks/K20-outer_prudent_family.py`: Exhausts six H parameters of size<=15; all 102 pinned fixed-point cells pass GLS with mutation controls; common F/P rows distinguish all six; explicit matching-G and ConBot/V separators. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K21

For m>=1, j>=1 (height semantics of closed GL fixed points), H_{m,j}=[]^{m+1}(p&(q|[]([]^jF|~q))) is unexploitable; DB trace h<=m; CC with F_k iff k<=j, P_i iff i>m; H vs

H defect (j<=m); S_j=[]^{j+1}(p|~[]^{j+1}q) gives D/C vs H_{m,j}, D/D vs G_{m,j}. Lean.

Status: derived.

Verified.

- `checks/K21-c046_hfam_lean.py`: embedded Lean 4 theorems h_fail, h_unexploitable, h_db, h_cb, h_conbot, h_fk, h_pj, h_hh, s_db, h_sep, g_sep, h_params compile with no sorryAx; 4 false variants rejected; literal height tables m,j<=8, k,i<=11, H-vs-H m,m'<=5, H/G vs S_j agree; independent string solver for m,j<=3: self, FB, CB, DB, ConBot, F_1..4, P_0..5, S_j; size(H)=m+j+10, size(S_j)=2j+6 — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K22

In L, the size-13 agent J=[][]([]p&(q|[]([]F|~q))) has self DD and FB/CB CC. Its F1..4/P0..5 outcomes match G21 and H21, but two size<=9 opponents distinguish all three; all 48 stated cells are GLS-certified.

Status: derived.

Verified.

- `checks/K22-mixed_prudent.py`: 48 hash-pinned GLS-certified cells with mutation controls; all three agents size13; named rows agree; two small opponents distinguish G21,J,H21. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K23

For $J_{a,b,j} = \Box^{a+1}(\Box^b p \wedge (q \vee \Box(\Box^j F \vee \neg q)))$, $0 \le a, b \le 5$, $1 \le j \le 6$: CC with F_k ($k \le 9$) iff $k \le j$, P_i ($i \le 9$) iff $i > a + b$; vs $S_{u,t}$ ($u, t \le 8$): CC if $u \le j$, else D/C iff $t + b \le u$, else DD. The 70 with $j \le a + b \le 5$ are distinct.

Status: derived.

Verified.

- `checks/K23-c046_jfam.py`: literal tables a,b<=5, j<=6: DB/CB/ConBot, F_k (k<=9), P_i (i<=9), S_{u,t} (u,t<=8) laws incl. defection heights, J vs J defect, random-trace safety; 70 members (j<=a+b<=5) have distinct vectors; independent string solver: 882 cells for all 21 members of size<=15 — pass

- `checks/K23-c046_jfam_gls.py`: all 21 J_{a,b,j} with 1<=j<=a+b, size<=15: 265 cells (self, CB, DB, ConBot, F_j/F_{j+1}, P_{a+b}/P_{a+b+1}, S_{j,j+3}, S_{u,t} boundaries t+b=u vs u+1 for u=j+1,j+2, G21/J111/H21 split on S_{2,1},S_{2,2}) match the K23 law formulas and pass GLS fixed-point certificates with flipped-candidate rejection — pass

Prior art. Possibly new: no prior art found in the owner's search.

Provenance. Run `20261017-1126`, claims K14, K18, K20, K21, K22, K23, run-folder commit `662166ecff0651902cc5fcb04bf747c62d795af2`, published 2026-10-18.

4 A certified map of variants of Gödel's ontological argument

Setting

Gödel's ontological argument, in Scott's version, uses a second-order predicate P ("positive") and five axioms: A1 (exactly one of a property and its negation is positive; split here into A1a, "not both", and A1b, "at least one"), A2 (positivity is closed under necessary entailment), A3 ($P(G)$, where G, "God-like", means having every positive property), A4 (positive properties are necessarily positive) and A5 (necessary existence is positive). The conclusion is that a God-like being exists, $\exists x\, G(x)$, and necessarily exists, $\Box\exists x\, G(x)$. A well-known side effect is *modal collapse*: everything true is necessarily true.

Many variants and weakenings are studied, mostly one variant at a time with an automated theorem prover [6, 4, 2, 5]. This finding maps the space systematically. We take every subset of each variant's axioms and the nine frame classes K, D, T, B, K4, KB, KD45, S4 and S5. We vary the semantics: all intensional properties, rigid properties or extensional properties (Fitting [11]), and constant or varying (actualist) domains. Cell by cell we decide whether the axioms are consistent, whether they prove the conclusion and whether they force collapse.

Result

- **Scott, all properties** (constant or varying domains): the God-like being exists exactly when the subset contains A1a, A2, A3 and A5 and the frames are B, KB or S5; necessary existence also holds on KD45. Collapse holds exactly when the subset contains A1a, A2 and A5, on the same frames. A1b and A4 are never needed. With A1, A2 and A5 alone, every symmetric frame is forced to consist of isolated worlds.
- **Scott, rigid properties:** with constant domains no consistent cell forces collapse, and the conclusion holds on every frame exactly when the subset

contains A1, A2 and A3, or A1, A3 and A5 (with Scott's essence). With varying domains existence needs A1a, A2, A3 and A5 on reflexive frames (T, B, S4, S5), or A4 as well on KB; with A4 added, necessary existence holds on every frame.

- **Anderson's variant** (with $H = P(G_A)$): existence needs A1a, A2 and H on B, KB or S5, and on KD45 only with A4 or A1b, and only with constant domains. A5 follows from A2 and H. Adding A1b, the direction Anderson removed, brings collapse back on B and S5, and on KB only together with A4.

- **Fitting's extensional variant** ($H = P(G)$, $N = P(NE)$): with constant domains no cell forces collapse, and the conclusion holds exactly for supersets of {A1a, A2, H} or {A1a, H, N}. With varying domains existence needs all of {A1a, A2, H, N} and reflexive frames (T, B, S4, S5); necessary existence also holds on KD45.

- **Benzmüller's simplified variant** (no positive empty property, closure under entailment, closure under arbitrary meets): every cell is consistent and none collapses; the conclusion holds exactly for the full axiom set with closure under local entailment, and never with closure under necessary entailment alone.

Why it matters

Each of these variants has been studied before, and many single cells of the map are known: the dependence of collapse on symmetry, the sufficiency of KB, the redundancy of A5 in Anderson's variant, the absence of collapse in Anderson's, Fitting's and the simplified variant [6, 4, 7, 2]. What we did not find is the complete picture: for each variant, exactly which axiom subsets and frame conditions give the conclusion or collapse, and the minimal sets. The map shows, for example, that in Scott's version neither A1b nor A4 ever matters, and that with rigid properties and constant domains collapse disappears altogether.

Proved vs checked

Every positive cell is proved in Lean 4 (core library, standard axioms only) in a shallow embedding of higher-order modal logic on arbitrary Kripke frames of the class. Every negative cell has an explicit finite countermodel, re-evaluated by a script. The tables cover the stated axiom sets, semantic choices and nine frame classes; other variants (other definitions of essence, other conjunction principles) are outside the map. We did not include the variants with Gödel's binary conjunction axiom: that it fails on infinite domains is known [3].

This result extends the single cells and per-proof axiom dependencies of Scott's and Gödel's variants established with other provers (Benzmüller & Scott 2025; Benzmüller 2026); the analysis of Anderson's variant and the redundancy

of its axioms (Benzmüller, Weber & Woltzenlogel Paleo 2017); Fitting's and Anderson's collapse-free variants (Fitting 2002; Benzmüller & Fuenmayor 2019); Benzmüller's simplified variant (2020). Our prior-art search (2026-10-19) covered:

- Benzmüller & Scott, "Notes on Gödel's and Scott's variants of the ontological argument", Monatshefte für Mathematik (2025), and its Isabelle/HOL development (AFP entry Notes_On_Goedels_Ontological_Argument, theory outline): per-theorem dependencies; modal collapse for Gödel variants 2 and 3 needs "only symmetry"; Scott's T3 and MC use Rsymm; Scott's variant fails in K only in the last step; UniqueEss lemmas (an essence of x is necessarily had only by x).

- Benzmüller, "Gödel's and Scott's Variants of the Ontological Argument in Lean 4 and TPTP THF" (arXiv 2609.26806, Oct 2026): Lean port with `#print axioms` dependency tables (Table 4) over S5, S4 and K embeddings, actualist/possibilist/mixed quantifiers; Th4 of Gödel variant 2 does not need Ax4; Scott's T3 and MC depend on symmetry alone; Gödel's 1970 axioms inconsistent in K.

- Benzmüller, "Proofs Without Nominals: Gödel's Ontological Argument, its Shallow Embedding, and the Open Questions of the Monatshefte Notes" (arXiv 2609.36279, 2026): the conjunction axiom must be emended (two or more conjuncts); collapse in K under the original axiom; in variant 3 an essence of x can only be "being x".

- Benzmüller & Woltzenlogel Paleo, automating Gödel's proof (20142016, incl. arXiv 1308.4526): KB suffices; Scott's axioms imply collapse; Gödel's original inconsistent.

- Benzmüller & Fuenmayor, positive properties, ultrafilters and modal collapse (arXiv 1910.08955): Anderson and Fitting variants avoid collapse.

- Benzmüller, "A (simplified) supreme being necessarily exists" (arXiv 2001.04701; KR 2020): simplified axioms valid in K and T without collapse.

- github.com/drewarrowood/godel-ontological (Lean 4): Scott with frame hypotheses; Symmetric gives local T3, S4 countermodel; rigid (world-independent) positivity makes A4 free but admits a countermodel to global T3; Russell-cut table (dropping A5 or A3).

- Fitting, Types, Tableaus and Gödel's God (2002); Sobel, Logic and Theism (2004); Anderson (1990); Wikipedia "Gödel's ontological proof" and "Modal collapse".

- Benzmüller, "A Comment on Modal Collapse and Ultrafilters in Gödel's Ontological Argument" (arXiv 2608.07578, 2026), replying to Odifreddi & Gomes, Logica Universalis (2026).

- Benzmüller, Weber & Woltzenlogel Paleo, "Computer-Assisted Analysis of the AndersonHájek Ontological Controversy" (Logica Universalis 2017; summary version read).
- Anderson & Gettings, "Gödel's Ontological Proof Revisited", in Gödel '96 (abstract only).

If you know of related work we missed, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was proved in Lean; the proof and check files are included so anyone can re-run them.

Claim K2

Full-property constant-domain embedding, Scott essence: A1, A2, A5 imply $R\,w\,v \leftrightarrow v = w$ and modal collapse on every symmetric frame; adding A3 gives $\exists x\, G(x)$ and $\Box\exists x\, G(x)$ there, and $\Box\exists x\, G(x)$ on euclidean frames. A4 is not used.

Status: derived.

Verified.

- `checks/K2-c009_lean_table.py`: lean –stdin exit 0, no sorry, no axiom decls; haec_ess, NE_isolated, A125_isolated_succ, A125_sym_iso, A125_sym_collapse, A1235_sym_god, A1235_euclid_box, majority_model all within {propext, Classical.choice, Quot.sound} — pass

Prior art. Partial prior art.

Claim K7

Constant domain, rigid properties, Scott/Gödel essence, all 32 subsets Œ 9 classes certified: no consistent cell forces collapse; $\exists x\, G(x)$ and $\Box\exists x\, G(x)$ hold iff the subset contains A1,A2,A3, or A1,A3,A5 with Scott essence, on every frame.

Status: derived.

Verified.

- `checks/K7-c009_rigid_table.py`: lean –stdin exit 0, no sorry, no axiom decls; A123_god/box, A135S_god/box, A12_A5S, A125G_no_world within {propext, Classical.choice, Quot.sound}, scott_rigid_noncollapse axiom-free; all 576 cells Œ 4 outcomes certified by a brute-force re-evaluated model or a Lean theorem whose hypotheses cell axioms; no pool model contradicts a Lean cell; stated pattern asserted for every cell — pass

Prior art. Partial prior art.

Claim K9

Constant-domain full-property Anderson slice: $\exists x\, G_A(x)$ holds iff the subset contains A1a, A2, H and the class is B, KB, S5, or KD45 with A4; $\Box\exists x\, G_A(x)$ iff it contains A1a, A2, H on B, KB, KD45, S5. A5 is derivable from A2 and H.

Status: derived.

Verified.

- `checks/K9-c009_anderson_open.py`: lean –stdin exit 0, no sorry, no axiom decls; NE_univ, A2H_A5, poss, sym_god, sym_box, eucl_box, eucl_A4_god, kd45_countermodel within {propext, Classical.choice, Quot.sound}; another researcher's verbatim evaluator gives the 3-world KD45 countermodel axiom mask {A1a,A2,H,A5}, no God-like being at the root, box-exists true; A2&H imply A5 on all 4136 models of K8's bound; resolves all 16 open K8 cells — pass

Prior art. Partial prior art.

Claim K13

Constant-domain extensional Fitting, all subsets of A1a,A1b,A2,H,N Œ nine frame classes: all 288 cells are consistent and permit noncollapse; both existence conclusions hold exactly for supersets of $\{A1a, A2, H\}$ or $\{A1a, H, N\}$. A1b is unnecessary.

Status: derived.

Verified.

- `checks/K13-fitting_split.py`: Lean core proves four existence/box theorems with A1a only and NE universal; standard axioms only, no sorry or axiom declarations. Re-evaluates explicit models covering every negative cell and all consistency/noncollapse cells; regenerates pinned 288-cell table. — pass

Prior art. Partial prior art.

Claim K15

Varying-domain (actualist) full-property Anderson, 32 subsets Œ 9 classes: as in constant domains except $\exists^E x\, G_A(x)$ on KD45 with A4 now fails; a 3-world KD45 model satisfies all five axioms, but its God-like being does not exist at the root.

Status: derived.

Verified.

- `checks/K15-c009_varying_anderson.py`: lean –stdin exit 0, no sorry, no axiom decls; god_exists_succ, poss, sym_god, sym_box, eucl_box, A2H_A5, kd45_varying, entails_top within {propext, Classical.choice, Quot.sound}; guarded evaluator = another researcher's verbatim evaluator at E=all on 4136 models; kd45_varying re-evaluated (mask 31, exists false, box-exists true, noncollapse); varying-E sweep of 16528 pairs consistent with all theorems; 288-cell table asserted (pattern, no OPEN, 15/KD45 and 31/KD45 refuted) and sha256-pinned — pass

Prior art. Partial prior art.

Claim K16

Varying-domain (actualist) extensional Fitting, 32 subsets of A1a,A1b,A2,H,N Œ 9 classes: $\exists^E x\, G(x)$ holds iff the subset contains $\{A1a, A2, H, N\}$ on T, B, S4, S5; $\Box\exists^E x\, G(x)$ also on KD45. K13's sets $\{A1a, A2, H\}$ and $\{A1a, H, N\}$ fail in S5.

Status: derived.

Verified.

- `checks/K16-c009_varying_fitting.py`: lean –stdin exit 0, no sorry, no axiom decls; god_nonempty, refl_god, T_god, T_box, eucl_box within {propext, Classical.choice, Quot.sound}; haec_ess, NE_iff, god_exists_succ, noN_S5, noA2_S5 axiom-free; 1134 search certificates re-evaluated by a direct evaluator per varying mode; table identical for possibly-empty, nonempty and every-individual-somewhere domains; const mode equals K13 pattern; no countermodel contradicts a Lean cell; 288-cell pattern asserted, no OPEN, sha256-pinned — pass

Prior art. Partial prior art.

Claim K18

Varying-domain rigid Scott, 64 subsets of A1a,A1b,A2..A5 Œ 9 classes: A1b is never needed; $\exists^E G$ iff {A1a,A2,A3,A5} on T,B,S4,S5 or +A4 on KB; $\Box\exists^E G$ iff {A1a,A2,A3,A5} on T,B,S4,S5,KD45 or +A4 on any frame.

Status: derived.

Verified.

- `checks/K18-c009_varying_rigid_split.py`: lean –stdin exit 0, no sorry, no axiom decls; god_nonempty, serial, T_god, T_box, eucl_box, A4_box, sym_A4_god (all with A1a only) within {propext, Classical.choice, Quot.sound};

haec_ess, NE_iff, god_exists_succ, A4_god_back, godel_no_world, kd45_A4, kb_noA4 axiom-free; 4274 search certificates re-evaluated by a direct evaluator; 1152-cell pattern asserted with no OPEN cell; table identical for nonempty domains; constant mode matches K7/K13 split pattern; sha256 70aa8973... — pass

- `checks/K18-c009_varying_rigid_split.py`: lean –stdin exit 0, no sorry, no axiom decls; 14 theorems within standard axioms; 4274 search certificates re-evaluated; 1152-cell pattern asserted with no OPEN cell; sha256 70aa8973... — pass

Prior art. Partial prior art.

Claim K21

Anderson G_A, full properties, 64 subsets of A1a,A1b,A2,H,A4,A5 Œ 9 classes Œ constant/varying domains: collapse iff {A1a,A1b,A2,H} on B,S5, or +A4 on KB; $\exists G_A$ on KD45 needs A4 or A1b and constant domains; otherwise as K9/K15.

Status: derived.

Verified.

- `checks/K21-c009_anderson_a1b.py`: lean –stdin exit 0, no sorry, no axiom decls; god_det, eucl_A1b_godlike, eucl_A4_godlike, refl_sym_collapse, sym_A4_collapse, sym_god, sym_box, eucl_box, A2H_A5, kb_cycle, kd45_A1b and 6 helpers within {propext, Classical.choice, Quot.sound}; 16528 bound models + 5 named models re-evaluated; no model contradicts a Lean cell; 1152-cell pattern asserted, all consistent, no OPEN; sha256 08d76079... — pass

Prior art. Partial prior art.

Claim K22

Full-property Scott, constant or varying domains, 64 subsets of A1a,A1b,A2A5 Œ 2 essences Œ 9 classes: A1b is never needed; $\exists G$ iff {A1a,A2,A3,A5} on B,KB,S5 (box also KD45); collapse iff {A1a,A2,A5} there; Gödel essence with {A1a,A2,A5} has no model.

Status: derived.

Verified.

- `checks/K22-c009_scott_split.py`: lean –stdin exit 0, no sorry, no axiom decls; serial, T1, A125_sym_collapse, A1235_sym_god, A1235_euclid_box, godel1970_no_model (hypotheses A1a only, E free) within {propext, Classical.choice, Quot.sound}; haec_ess, entails_top axiom-free; K3 data sha256

91c8ed5b... pinned; 3116 model outcomes re-evaluated (class, split axioms, refuted outcome) incl. the all-positive model; 864-cell pattern asserted, A1b-drop invariance asserted, table sha256 eb8f629f... pinned — pass

Prior art. Partial prior art.

Claim K24

Constant-domain full-property simplified AllMeet slice (216 cells): every cell is consistent and permits noncollapse; $\exists G$ and $\Box\exists G$ hold exactly for all three axioms with local/either closure. Every other cell has a finite countermodel.

Status: derived.

Verified.

- `checks/K24-allmeet_classification.py`: Self-contained Lean existence/box proofs plus regeneration of SHA256-pinned 216-cell AllMeet table; every finite certificate re-evaluated; arbitrary meets checked exactly by top and binary intersections on finite powersets; no open cells. — pass

Prior art. Partial prior art.

Claim K28

Varying-domain full-property simplified slices: AllMeet has exactly K24s 216-cell outcomes; binary closure has 216 consistent, noncollapsing cells refuting $\exists^E G$ and $\Box\exists^E G$, even with A1w. Positive AllMeet cells require all axioms and local/either closure.

Status: derived.

Verified.

- `checks/K28-varying_meets.py`: Lean core with arbitrary E proves AllMeet PG/existence/box under A1prime and guarded local closure, plus either-to-local; guarded total-E tail model refutes existence/box and inherited proofs certify binary closure/A1w/noncollapse. Regenerates SHA256-pinned 216-cell AllMeet table and reevaluates every finite certificate with explicit E; enumerates 216 binary rows by weakening the Lean infinite model; no sorry or declared axioms. — pass

Prior art. Partial prior art.

Discussion

The map is a refinement of work that is largely known, not a new argument. Its value is completeness: for each variant, every subset of the axioms and every listed frame class is decided with a certificate, so the minimal sets and the frame boundaries can be read off rather than inferred from separate studies. Several entries agree with results already established with other provers [6, 4, 7], which we take as a check on our encoding rather than as findings.

The map covers only the stated choices: a shallow embedding of higher-order modal logic, Scott's and Gödel's definitions of essence, the listed variants, nine frame classes and two kinds of domain. Other definitions of essence, other readings of positivity (for example world-relative and rigid mixtures), Gödel's conjunction axiom in all slices [3], and logics beyond these frame classes are outside it. A cell decided here can therefore change under a different encoding, and claims of minimality hold relative to the listed axioms.

What a reader can take away is where the strength of each variant sits: in Scott's version the weight lies on A1a, A2, A3 and A5 together with symmetry, while A1b and A4 are idle; in Anderson's version the dropped complement direction is exactly the switch for collapse; and in the rigid and simplified readings, existence holds without collapse. Natural next steps are the open cells the run left (mixed conjunction principles in some slices) and other definitions of essence.

Provenance. Run `20261018-1933`, claims K2, K7, K9, K13, K15, K16, K18, K21, K22, K24, K28, run-folder commit `338871c09385f72bc0f657c9a639ef6a3c825566`, published 2026-10-19.

5 A census of modal agents in the prisoner's dilemma

Setting

Two programs play a one-shot prisoner's dilemma, and each can read the other's source code. Following Barasz et al. [1], a *modal agent* decides by asking what can be proved about the opponent, written with the provability box $\Box$ of the logic GL: p means "the opponent cooperates with me", q "the opponent cooperates with DefectBot". We take the language L of all formulas built from p, q, $\bot$, $\neg$, $\wedge$, $\vee$ and $\Box$ with every atom under a box, measure size as the number of nodes, and identify agents that behave identically against every opponent. Payoffs are the standard 5 (temptation), 3 (mutual cooperation), 1 (mutual defection), 0 (sucker). An earlier finding on this site describes the self-defecting families found in this census; this one reports the census itself.

Result

- **Counts.** Up to size 9 there are 2815 behaviourally distinct agents (new per size 1 to 9: 1, 4, 8, 12, 31, 98, 255, 646, 1760), and every pairwise outcome is solved exactly.
- **No evolutionarily stable agent.** Up to size 9, 180 of the 1335 observational classes are symmetric Nash equilibria, but none is an ESS. Every self-defecting equilibrium is invaded by FairBot, an argument that holds for every modal agent, not only those in the census. Under replicator dynamics from the uniform start, every surviving agent cooperates with itself and the mean payoff tends to 3.
- **The round-robin winner never cooperates.** Up to size 9 the top score belongs to agents that act exactly like DefectBot. Up to size 11 the top 16 agents defect against everyone yet beat DefectBot, because more opponents cooperate with them (9863 against 9688); for example $\Box\Box p \wedge \neg\Box(p \vee \Box\bot)$.
- **Self-defecting cooperators are almost always exploitable.** Among agents that defect against themselves but cooperate with FairBot and CooperateBot (2, 2, 4, , 2468, 7032 per size 4 to 14), only 3, 19 and 75 at sizes 12, 13 and 14 are unexploitable by every modal opponent. Each of the others is exploited by one of four simple agents: $\Box\neg p$, $\Box p$, $\Box q$ or $\Box\bot$.
- **A threshold law, and where it breaks.** Up to size 14, every self-defecting agent that cooperates with FairBot cooperates with graded FairBots $F_k = \Box^k p$ exactly for $k \leq a$ and with graded PrudentBots P_i exactly for $i \geq b$, for some $a < b$. The law fails at size 17: the agent $Y = \Box(\Box\Box\bot \vee (p \wedge (q \vee \neg\Box(q \wedge \Box\Box\Box\bot))))$ cooperates with F_1 and F_k for $k \geq 4$, but not with F_2 or F_3.

Why it matters

The literature on modal agents studies a handful of named agents (FairBot, PrudentBot, CliqueBot and a few more) [1, 14]. A census makes the space itself visible: it shows that program equilibrium among modal agents has no evolutionarily stable point, that a round robin rewards agents that never cooperate, and that self-defecting cooperators are rarely safe, with four tiny agents enough to expose all the others.

Proved vs checked

Every outcome is computed exactly in the height semantics of closed GL fixed points (see [8]; the link to provability in arithmetic is Solovay's theorem [17]) and re-checked on large samples by an independent solver; the datasets are pinned by hash. The counts, equilibria and exploitability results hold for all agents

up to the stated sizes, not in general. The FairBot invasion argument and the counterexample Y hold in general.

We believe this result is new. Our prior-art searches (2026-10-17 to 2026-10-19) covered:

- OEIS (oeis.org), searched for the census counts 1, 4, 8, 12, 31, 98, 255, 646, 1760 (per size) and 1, 5, 13, 25, 56, 154, 409, 1055, 2815 (cumulative): no match.
- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; LessWrong / Arbital "Modal combat"; Mennen 2017; Critch, Dennis & Russell 2022 (arXiv 2208.07006); github.com/machine-intelligence/provability.
- Barasz et al. 2014, Robust Cooperation in the Prisoner's Dilemma (arXiv 1401.5577), full text: named agents CB, DB, CliqueBot, FairBot, PrudentBot, TrollBot, JustBot, WaitFairBot_K; Theorems 3.2, 4.8, 4.10; obstacles section.
- LaVictoire et al. 2014, Program Equilibrium in the Prisoner's Dilemma via Löb's Theorem.
- LessWrong post on Barasz et al. and its comments (orthonormal, cousin_it, Will_Sawin, Karl, dankane).
- Arbital / LessWrong wiki "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling"; Garrabrant, "Modal SAT: Introduction" (2015).
- Mennen, "Modal Combat for games other than the prisoner's dilemma".
- Critch, Parametric Bounded Löb's Theorem and Robust Cooperation of Bounded Agents (arXiv 1602.04184), first sections.
- Searches for round-robin, tournament, replicator or ESS analyses of modal agents: only iterated-PD and FDT evolution papers (e.g. arXiv 2005.05154, no modal agents).
- github.com/machine-intelligence/provability (agents folder contents not visible).
- Barasz et al. 2014 (arXiv 1401.5577), full text; LaVictoire et al. 2014 (Program Equilibrium via Löb's Theorem).
- LessWrong post on Barasz et al. and comments; Arbital "Modal combat".
- Garrabrant, "Prisoners' Dilemma with Costs to Modeling" and "Modal SAT: Introduction"; Mennen, "Modal Combat for games other than the prisoner's dilemma".
- Critch, Parametric Bounded Löb (arXiv 1602.04184), first sections.

- Oesterheld 2022, A Note on the Compatibility of Different Robust Program Equilibria of the Prisoner's Dilemma (arXiv 2211.05057).

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was checked by script; the proof and check files are included so anyone can re-run them.

Claim K1

`In L (atoms p,q each under a box; F,~,&,|,[]; size=node count) agents up to size 9 form 2815 classes of equivalence on finite linear GL models (new per size:  1,4,8,12,31,98,255,646,1760); every pairwise outcome is solved exactly.`

Status: derived.

Verified.

- `checks/K1-c046_census9.py`: class counts per size 1..9; pinned dataset equals rebuilt class list; 18890 games (6 named agents' full rows/columns + 2000 random) agree with an independent height-table solver — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K12

`In L up to size 9 (2815 classes, payoffs 5/3/1/0):  the round-robin maximum 8083 is reached only by 63 agents acting exactly as DefectBot; of 1335 behaviour classes 180 are symmetric Nash (78 self-defecting, all invaded by FairBot) and none is an ESS.`

Status: derived.

Verified.

- `checks/K12-c046_structure9.py`: sha-pinned census9 matrix: round-robin max 8083 by exactly 63 DB-identical agents, second 7979; 1335 observational classes; 180 Nash (78 defect, 102 coop); FB invades all 78; each of 102 has a neutral mutant (CB for 95); no ESS; 183965 used/random games re-solved by independent height-table solver — pass
- `checks/K12-c046_replicator9.py`: pinned census9; discrete replicator, uniform over 2815 chain classes, 20000 steps (exact 1335-class reduction verified 300 steps <1e-12): mean payoff 3 within 1e-6, 788 survivors >1e-6, all self-cooperating, 0.38+-0.005 weight on exploitable classes, final vector equals embedded pin within 2e-6 — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K15

`In L up to size 11 (20441 classes, payoffs 5/3/1/0): round-robin max 59893 is reached only by 16 agents of size 10-11 that defect against all, e.g. []p&~[](p|[]F); DefectBot scores 59193 (rank 21). 1120 symmetric Nash classes, no ESS.`

Status: derived.

Verified.

- `checks/K15-c046_structure11.py`: sha-pinned census11 agents+matrix: rr max 59893 by exactly 16 total defectors of sizes 10,11, each cooperated with by 9863; DB 59193, 9688, 20 above; T vs []F|~[]~p = D/C; 9095 obs classes; Nash 1120 (401 FB-invaded, 719 with neutral mutant); ESS none; 53938 cells re-solved independently — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K17

`In L up to size 14, of the agents that defect vs themselves but cooperate with FairBot and CooperateBot (2,2,4,...,2468,7032 per size 4..14), 3/19/75 at sizes 12/13/14 are unexploitable by every modal opponent; each other is exploited by []~p, []p, []q or []F.`

Status: derived.

Verified.

- `checks/K17-c046_dichotomy14.py`: census counts to size 14 (..,36062,100823,279115); combination counts per size 4..14; machine criterion U holds for exactly 3,19,75 members at sizes 12,13,14 (sha256-pinned list incl. W) and none below; every non-U member exploited by one of []~p,[]p,[]q,[]F; independent height-table solver re-checks 400 witnesses and all 97 U agents' properties; random-trace test finds no exploit for U agents and finds one for 40/40 non-U controls — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K19

`In L up to size 14 (and 15, unchecked run), every self-defecting agent cooperating with FairBot cooperates with F_k (k<=8) iff k<=a and with P_i iff i>=b, a<b. Fails at size 17: Y=[]([][]F|(p&(q|~[](q&[][][]F)))) coops F_1, F_k k>=4, P_i i>=3, not F_2,F_3.`

Status: derived.

Verified.

- `checks/K19-c046_segment.py`: census counts to 14; law (initial F-segment, final P-segment, a<b) on all 9979 S=0 FB-cooperating classes; Y size 17, S=0, CC FB/CB/F_1/F_4..F_12/P_3..P_12, DD with F_2,F_3,P_0..P_2, D vs DB, criterion U; independent string solver for all Y rows; 3000 random traces; GLS certificates with flipped controls for Y vs Y,F_1..F_4,P_2,P_3,CB — pass

Prior art. Possibly new: no prior art found in the owner's search.

Discussion

The census is exhaustive only up to the stated sizes (9 for the counts and equilibria, 11 for the round robin, 14 for exploitability and the threshold law), and only for the language L with the single probe "the opponent cooperates with DefectBot". Richer probes (how the opponent treats FairBot or CooperateBot) would give a different, larger space, and the counts depend on the exact choice of connectives and size measure. Results about tournaments and replicator dynamics depend on the payoffs and on the uniform starting population; we report them as properties of this population, not of modal agents in general. The FairBot invasion argument and the counterexample Y are the exceptions: they hold for all modal agents.

Two questions stay open. First, whether the unexploitable self-defecting agents found up to size 14 all belong to known families: some of them do, and the rest were shown different from family members of small index but not classified. Second, whether the threshold law, which fails at size 17, holds in a weaker form, for example for all agents of a given family. The natural next searches are the same census for other two-action games (Chicken, Stag Hunt), where outcomes are unchanged and only the scoring differs, and a census with more probes.

Provenance. Run `20261017-1126`, claims K1, K12, K15, K17, K19, runfolder commit `662166ecff0651902cc5fcb04bf747c62d795af2`, published 2026-10-19.

6 Settling informal claims about modal agents in Chicken

Setting

Modal agents play one-shot games by asking what can be proved about each other, written with the provability box $\Box$ of the logic GL [1]. Two informal posts made claims about specific agents that were never checked. "Modal Chicken" [12] introduced CarefulBot and AstuteBot for the game of Chicken,

and its comments proposed graded variants: $C_n = \neg\Box(\Box^n\bot \vee p)$ ("swerve unless the opponent provably swerves", with proof strength n) and BoldBots $B_n = \Box(\Box^n\bot \vee \neg p)$, the name a commenter gave to the post's original CarefulBot (B_0); here p means "the opponent swerves against me". "Yet More Modal Combat" [13] proposed, for the prisoner's dilemma, a bot that defects if a weak system proves the opponent cooperates and otherwise acts like FairBot with a strong system; in graded form $H_{a,b} = \neg P_a(p) \wedge P_b(p)$ with $P_a(z) = \Box(\Box^a\bot \vee z)$, $a < b$.

Result

- **Graded CarefulBots and BoldBots, for all grades:** C_n swerves against C_m exactly when $n \leq m$, while B_n swerves against B_m exactly when $n > m$. A CarefulBot against a BoldBot always goes straight while the BoldBot swerves, so the pair always ends at payoffs $(4, 1)$, whatever the grades.

- **The named agents:** a certified table of all 64 pairings of SwerveBot, NerveBot, the original CarefulBot B_0, C_0, AstuteBot as displayed in the post, the FairBots $P_0(p) = \Box p$ and $P_1(p)$, and Hobson's bot $H_{0,1}$ (which cooperates with itself but ends in mutual defection with both FairBots). The original CarefulBot crashes against itself, as a comment pointed out; the displayed AstuteBot swerves against itself, as the post claimed.

- **Hobson's bot, for all grades:** $H_{a,b}$ and $H_{c,d}$ end in mutual cooperation exactly when their weak grades agree ($a = c$), and otherwise in mutual defection. Against any FairBot of the form $P_e(p)$ it ends in mutual defection, for every e.

Why it matters

Informal claims about small agents are easy to make and easy to get wrong; two of the claims in these posts were already disputed in their comment threads. These results settle them for every grade at once. They also show that Hobson's bot, proposed as a bot that cooperates with FairBots, does not cooperate with any graded FairBot of this form.

Proved vs checked

The graded laws (CarefulBots, BoldBots, the mixed pair, Hobson's bot) are proved for all indices in Lean 4, in the height semantics of closed GL fixed points (see [8]; [17]). The 64-cell table is certified by GL fixed-point computation.

This result extends the informal claims about CarefulBot, AstuteBot and graded CarefulBots and BoldBots in "Modal Chicken" and its comments (Gram_Stone 2015); Hobson's two-system bot and the counterexample in its comments (Hobson 2021). Our prior-art search (2026-10-20) covered:

- Mennen, "Modal combat for games other than the prisoner's dilemma" (LessWrong / Alignment Forum, 2017): framework for modal agents in games with any number of players and actions (player types, ranks, opponent tests naming a target player, an agent to test against and an action); existence and Pareto theorems; no agent families with index laws, no enumeration.

- Gram_Stone, "Modal Chicken" (LessWrong, 2015) and its comments (Alex-Mennen, SilentCal): SwerveBot, NerveBot, CarefulBot, AstuteBot; informal claims about graded CarefulBots and BoldBots.

- Hobson, "Yet More Modal Combat" (LessWrong / Alignment Forum, 2021): a two-system PD bot; generalisation to Chicken and more players left open.

- Critch, Dennis & Russell, "Cooperative and uncooperative institution designs: surprises and problems in open-source game theory" (arXiv 2208.07006): two-player bounded proof-search agents.

- Todorov, ten Napel & Müller, "Parametric Open Source Games" (arXiv 2606.27068): non-modal parametric agents; Chicken and Stag Hunt thresholds.

- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; Critch, parametric bounded Löb (arXiv 1602.04184); Oesterheld 2022 (arXiv 2211.05057); LessWrong / Alignment Forum wiki "Modal combat"; web searches for modal or provability-based agents in Chicken, Stag Hunt and multiplayer program equilibrium.

If you know of related work we missed, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was partly proved in Lean (claims K5, K11, K12) and partly checked by script (claims K6); the proof and check files are included so anyone can re-run them.

Claim K5

For all $n, m \geq 0$, $C_n = \neg P_n(p)$ swerves against C_m iff $n \leq m$, while $B_n = P_n(\neg p)$ swerves against B_m iff $n > m$, where $P_n(a) = \Box(\Box^n \bot \vee a)$. Exact height laws are Lean-proved.

Status: derived.

Verified.

- `checks/K5-informal_tables.py`: 50 graded pair outcomes for indices 0..4, each with both simultaneous GL fixed-point equations certified. — pass

- `checks/K5-graded_families.py`: Lean core proves exact height laws for every n,m,h and every solution, plus boxed-formula bridge; no sorryAx; false tie law rejected; 20480 finite recurrence cells agree. — pass (Lean-proved, no sorryAx)

Prior art. Partial prior art.

Claim K6

The 64-cell GL table for $S, N, B_0, C_0, A, H, F_0, F_1$ is certified: original CarefulBot B_0 self-crashes; displayed AstuteBot A self-swerves; $H = \neg\Box p \wedge \Box(\Box\bot \vee p)$ self-cooperates but mutually defects with F_0, F_1.

Status: derived.

Verified.

- `checks/K5-informal_tables.py`: All 64 stated outcomes, both GL fixed-point equations per cell, rejection of flipped candidates, and closed-form tail evaluation. — pass
- `checks/K5-informal_tables.py`: 64 outcomes and both simultaneous GL equations per pair, flipped controls rejected; no calculation changed. — pass

Prior art. Partial prior art.

Claim K11

For all $n, m \geq 0$, $C_n = \neg P_n(p)$ goes straight and $B_m = P_m(\neg p)$ swerves against it at every height, where $P_k(a) = \Box(\Box^k\bot \vee a)$. The mixed pair always yields Chicken payoffs $(4, 1)$, independent of grades (Lean).

Status: derived.

Verified.

- `checks/K11-mixed_graded.py`: Lean proves mixed_all for all natural indices/heights and every solution, with no axioms; boxed-formula bridge; 25 GLS cells and 10240 recurrence cells pass; false conclusion rejected. — pass

Prior art. Partial prior art.

Claim K12

For $H_{a,b} = \neg P_a(p) \wedge P_b(p)$, $P_a(z) = \Box(\Box^a \bot \vee z)$, all $a < b, c < d$ give eventual mutual action 1 iff $a = c$, else mutual action 2. Every pairing with $P_e(p)$ eventually gives mutual action 2, for all e (Lean).

Status: derived.

Verified.

- `checks/K12-hobson_grades.py`: 100 generalized Hobson pair cells plus 50 graded FairBot cells; both fixed-point equations GL-proved per cell, flipped candidates rejected. — pass
- `checks/K12-hobson_all_indices.py`: Copied another researcher proof: Lean hob_eq, hob_tail, hob_fair for all indices, boxed bridge, no sorryAx; two false variants rejected; 7098 recurrence cells. — pass (Lean-proved, no sorryAx)

Prior art. Partial prior art.

Discussion

The results depend on reading each informal definition as a specific modal formula; where a post was ambiguous we state the formula we used, and a different reading could change the outcome. Hobson's proposal allows other pairs of systems than the graded ones here, so the negative result covers this graded family, not every possible instantiation.

Provenance. Run `20261019-1400`, claims K5, K6, K11, K12, run-folder commit `24c2a64bf8b8cb8be184b4d449ce11068d8270ea`, published 2026-10-20.

7 Three-player modal agents in a public-goods game

Setting

Modal agents decide what to do by asking what can be proved about the other players, written with the provability box $\Box$ of the logic GL [1]. Mennen [15] set up a framework for more than two players; here we study three players concretely. The game is a public-goods game: each player contributes ($a_i = 1$) or not, and earns $u_i = 2\sum_j a_j - 3a_i$, so contributing costs 1 net but gives each other player 2. An agent sees p_1, p_2 ("opponent j contributes in this game") and probes q_1, q_2 ("opponent j contributes when I am replaced by a defector"). Behaviour is read off in the long run (tail semantics). An agent is *safe* if it never earns less than either opponent, whoever they are.

Result

- **The three-player FairBot is safe.** $F = \Box(p_1 \wedge p_2)$ (contribute iff it is provable that both others contribute) never earns less than either opponent against any two modal opponents, contributes with two copies of itself and defects against a defector.
- **It is the smallest.** No agent smaller than F (size 4) both cooperates with copies of itself and is safe.
- **All small safe cooperators are FairBot-like.** Exactly 66 agents up to size 6 cooperate with themselves and are safe; none of them uses the probes, and for each GL proves that it contributes only if $\Box^3(p_1 \wedge p_2)$. Every other agent of that size fails against some pair of opponents of size at most 3.
- **A prudent three-player agent needs the probe.** $E = \Box((p_1 \wedge p_2) \wedge (\Box\bot \vee \neg q_1))$ (size 10) cooperates with itself, is safe, and defects against two unconditional contributors; no agent of size at most 7 has all three properties.
- **The definition of the probe matters.** For five basic three-player bots we certify all 125 triples. Changing the probe from "only I am replaced by a defector" to "everyone except the queried opponent is replaced" changes the outcome of 9 of the 125 triples.

Why it matters

Program equilibrium with modal agents has been worked out almost only for two players. In a three-player public-goods game the natural FairBot generalisation stays safe, is the smallest safe cooperator, and is essentially the only kind up to size 6. Exploiting unconditional contributors requires the probe, as with PrudentBot in two-player games, and the exact definition of the probe is a real modelling choice, since outcomes change with it.

Proved vs checked

The safety of F against all modal opponents is proved in Lean 4 under tail semantics. The minimality, the 66-agent classification, the properties of E and the 125-triple tables are exact computations over all agents up to the stated sizes in the stated language, with GL certificates.

We believe this result is new. Our prior-art search (2026-10-20) covered:

- Mennen, "Modal combat for games other than the prisoner's dilemma" (LessWrong / Alignment Forum, 2017): framework for modal agents in games with any number of players and actions (player types, ranks, opponent tests naming a target player, an agent to test against and an action); existence and Pareto theorems; no agent families with index laws, no enumeration.

- Gram_Stone, "Modal Chicken" (LessWrong, 2015) and its comments (Alex-Mennen, SilentCal): SwerveBot, NerveBot, CarefulBot, AstuteBot; informal claims about graded CarefulBots and BoldBots.
- Hobson, "Yet More Modal Combat" (LessWrong / Alignment Forum, 2021): a two-system PD bot; generalisation to Chicken and more players left open.
- Critch, Dennis & Russell, "Cooperative and uncooperative institution designs: surprises and problems in open-source game theory" (arXiv 2208.07006): two-player bounded proof-search agents.
- Todorov, ten Napel & Müller, "Parametric Open Source Games" (arXiv 2606.27068): non-modal parametric agents; Chicken and Stag Hunt thresholds.
- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; Critch, parametric bounded Löb (arXiv 1602.04184); Oesterheld 2022 (arXiv 2211.05057); LessWrong / Alignment Forum wiki "Modal combat"; web searches for modal or provability-based agents in Chicken, Stag Hunt and multiplayer program equilibrium.

We found no earlier statement of it, but we can't be certain. If you know of prior work, please open an issue.

This result extends Mennen's framework for multiplayer modal agents, whose opponent tests fix one definition of probe atoms (Mennen 2017). Our prior-art search (2026-10-20) covered:

- Mennen, "Modal combat for games other than the prisoner's dilemma" (LessWrong / Alignment Forum, 2017): framework for modal agents in games with any number of players and actions (player types, ranks, opponent tests naming a target player, an agent to test against and an action); existence and Pareto theorems; no agent families with index laws, no enumeration.
- Gram_Stone, "Modal Chicken" (LessWrong, 2015) and its comments (Alex-Mennen, SilentCal): SwerveBot, NerveBot, CarefulBot, AstuteBot; informal claims about graded CarefulBots and BoldBots.
- Hobson, "Yet More Modal Combat" (LessWrong / Alignment Forum, 2021): a two-system PD bot; generalisation to Chicken and more players left open.
- Critch, Dennis & Russell, "Cooperative and uncooperative institution designs: surprises and problems in open-source game theory" (arXiv 2208.07006): two-player bounded proof-search agents.
- Todorov, ten Napel & Müller, "Parametric Open Source Games" (arXiv 2606.27068): non-modal parametric agents; Chicken and Stag Hunt thresholds.

- Barasz et al. 2014 (arXiv 1401.5577); LaVictoire et al. 2014; Critch, parametric bounded Löb (arXiv 1602.04184); Oesterheld 2022 (arXiv 2211.05057); LessWrong / Alignment Forum wiki "Modal combat"; web searches for modal or provability-based agents in Chicken, Stag Hunt and multiplayer program equilibrium.

If you know of related work we missed, please open an issue.

This result was found by AI research agents (the autoresearch harness). It was partly proved in Lean (claims K14, K16, K18, K19) and partly checked by script (claims K13, K21); the proof and check files are included so anyone can re-run them.

Claim K13

For five three-player bots $D, C, F = \Box(p_1 \wedge p_2), O = \Box(p_1 \vee p_2), Q = \Box(q_1 \wedge q_2)$, 125 triples are GL-certified under evaluator-replacement probes: $FFF = 111$, $FFD = 000$, $QQQ = 000$; exactly D, F, Q never earn less than either opponent within this five-bot set.

Status: derived.

Verified.

- `checks/K13-three_player.py`: 125 triples, all 3000 simultaneous replacement-subset equations GL-certified; 375 flipped-candidate controls; headline cells and finite payoff/self-play classifications. — pass
- `checks/K13-probe_semantics_alternative.py`: Re-certifies the original 125 triples and alternative 125 triples; 6000 equations and 750 flip controls; exactly nine stated differences. — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K14

In the three-player public-goods game $u_i = 2\sum_j a_j - 3a_i$, $F = \Box(p_1 \wedge p_2)$ never earns less than either opponent against any two modal opponents (Lean, tail semantics), while cooperating with two copies and defecting against a defector.

Status: derived.

Verified.

- `checks/K14-c046_xreview_k13.py`: (4) Lean 4 core fair3_tail (any opponent traces, X->Y&Z) and fair3_payoff (decided over 8 tails), no sorryAx, false variant rejected; (2) FFF=111, FFD=000, QQQ=000, FCD=010, QBC=101 GL-certified on all 24 replacement equations; (1) 125 K13 triples re-solved independently — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K16

In the guarded three-player language with node-count size and evaluator-replacement probes, size 4 is minimal for self-cooperation plus never earning less than either opponent in $u_i = 2\sum_j a_j - 3a_i$: $F = \Box(p_1 \wedge p_2)$ attains it (tail semantics).

Status: derived.

Verified.

- `checks/K16-three_player_minimal.py`: Exhaustive 25 raw guarded trees through size 3; 15625 augmented triples, 375000 GL equations, 46875 flip controls; each smaller self-cooperator has a disadvantage witness; size-4 FairBot self-cooperates and passes finite safety. — pass
- `checks/K16-three_player_minimal.py`: 25 exhaustive raw trees through size 3; 15625 triples / 375000 GL equations / 46875 flip controls; smaller self-cooperators have disadvantage witnesses; embedded K14 Lean fair3_tail and fair3_payoff prove universal upper bound, no sorryAx, false variant rejected. — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K18

In the guarded three-player language (evaluator-replacement probes, $u_i = 2\sum_j a_j - 3a_i$), exactly 66 trees of size ≤ 6 self-cooperate and never earn less than either opponent; all are q-free and GL proves $A \to \Box^3(p_1 \wedge p_2)$; others fail vs size-$\leq 3$ pairs.

Status: derived.

Verified.

- `checks/K18-c046_safe6.py`: (1) 1,6,18,112,596,3756 guarded trees; (2) self-cooperators 11,41,263,1448; survivors 0,0,0,2,8,56, all others beaten by size<=3 pairs (constants except 8 named trees); (3) survivors q-free, GL proves A->[][][](p1&p2), control rejected; (4) 3438 triples GL-certified, 24 equations each, flips rejected; (5) Lean lag_tail, fair3_payoff, no sorryAx, false variant rejected — pass (Lean-proved, no sorryAx)

Prior art. Possibly new: no prior art found in the owner's search.

Claim K19

In K18's three-player language, $E = \Box((p_1 \wedge p_2) \wedge (\Box\bot \vee \neg q_1))$ (size 10) self-cooperates, defects against two constant contributors (payoffs 4,1,1) and never earns less than either opponent; no tree of size ≤ 7 has all three properties.

Status: derived.

Verified.

- `checks/K19-c046_prudent3.py`: (1) 9 E-cells (EEE=111, ECC=011, ECD=010, EDD=000, E FB FB=111, E E FB=111, E C FB=010, E D FB=000, EEC=111) GL-certified, 24 equations each, flips rejected; (2) GL proves E->[](p1&p2), control rejected; (3) sizes<=7: 1,6,18,112,596,3756,23636 trees, self-coop & defect vs (C,C) 0,0,4,4,68,220,1808, each beaten by a size<=3 pair; (5) Lean lag_tail, fair3_payoff, no sorryAx, false variant rejected — pass (Lean-proved, no sorryAx)
- `checks/K19-c046_xreview_k20.py`: K20 cells incl. RRR=000 (q1->p1) GL-certified on 24 equations, flips rejected; GL proves A<->E with 8 mixed triples 111; GL rejects A<->R (probe essential) — pass

Prior art. Possibly new: no prior art found in the owner's search.

Claim K21

For K13s five bots, replacing the evaluator alone versus replacing all players other than the queried opponent changes exactly 9 of 125 triples: permutations of (Q, C, O) and (Q, O, O) change from 111 to only Q defecting (GL-certified).

Status: derived.

Verified.

- `checks/K13-probe_semantics_alternative.py`: 250 triples, 6000 GL equations, 750 flips; exactly the specified nine differences, QOC 111 versus 011; all probe-free triples unchanged. — pass

Prior art. Partial prior art.

Discussion

The results use one language (two "contributes" atoms and two probes), one payoff function and tail semantics; other probe definitions change some outcomes, as shown. The classifications stop at sizes 6 to 10. Open questions are the minimum size of a safe self-cooperator that needs the probe (between 8 and 10) and whether these patterns extend to more players or other multiplayer games.

Provenance. Run `20261019-1400`, claims K13, K14, K16, K18, K19, K21, run-folder commit `24c2a64bf8b8cb8be184b4d449ce11068d8270ea`, published 2026-10-20.

8 How to verify

Each finding has its check scripts in `claims/<slug>/checks/` of the findings repository. Run each with `python -I <script>`; it passes if it exits with status 0. The checks are Python scripts and need the packages of the research image (the image's `requirements.txt`); the Lean checks also need `lean`.

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[17] Robert M. Solovay. Provability interpretations of modal logic. *Israel Journal of Mathematics*, 25:287–304, 1976.